A Coordinated Model Predictive Control-Based Approach for Vehicle-to-Grid Scheduling Considering Range Anxiety and Battery Degradation

Chuan-Fan Lu, *Graduate Student Member, IEEE*, Guo-Ping Liu, *Fellow, IEEE*, Yi Yu, *Member, IEEE*, and Jinqiang Cui

***Abstract*— Electric vehicles (EVs) spend the majority of their operational lifecycle in a state of nonuse, which provides sufficient time and flexibility for charging. Utilizing this flexibility with vehicle-to-grid (V2G) technology not only supports grid stability but also offers additional revenue for vehicle owners. However, V2G scheduling based solely on maximizing the individual profits of EVs will lead to a high cost of peak demand charge. Moreover, concerns about range anxiety, battery degradation, and the computational burden of optimization have constrained the widespread of V2G technology. To address these challenges, a coordinated model predictive control (CMPC)-based scheduling scheme is proposed while considering users' range anxiety and battery degradation model. A mixed integer linear programming (MILP) is developed to consider range anxiety, battery degradation model, and constant current–constant voltage (CC–CV) charging mode. The CMPC method is proposed to avoid the penalty of peak demand charge by coordinating V2G charging among chargers and to mitigate sharp fluctuations in power by adding a penalty term in the power change rate. Instead of optimizing calculations for each time stage, a control package strategy and an online rolling optimization method are presented to reduce computational burden and consider stochastic charging demand arrivals. Simulations and hardware experiments are performed to demonstrate the effectiveness and feasibility of the proposed scheme.**

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Chuan-Fan Lu is with the Center for Control Science and Technology, Southern University of Science and Technology, Shenzhen 518055, China, and also with Pengcheng Laboratory, Shenzhen 518108, China (e-mail: lucf2023@mail.sustech.edu.cn).

Guo-Ping Liu is with the Center for Control Science and Technology, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: liugp@sustech.edu.cn).

Yi Yu is with the Center for Control Science and Technology, Southern University of Science and Technology, Shenzhen 518055, China, and also with the Department of Electrical and Electronic Engineering, The Hong Kong Polytechnic University, Hong Kong, SAR, China (e-mail: yuyi@sustech.edu.cn).

Jinqiang Cui is with Pengcheng Laboratory, Shenzhen 518108, China (e-mail: cuijq@pcl.ac.cn).

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NOMENCLATURE

ac	Alternating current.
Ah	Ampere-hours.
CC	Constant current.
CC–CV	CC–constant voltage.
CMPC	Coordinated model predictive control.
CV	Constant voltage.
dc	Direct current.
EV	Electric vehicle.
IoT	Internet of Things.
MILP	Mixed integer linear programming.
MPC	Model predictive control.
SOC	State-of-charge.
SOE	State-of-energy.
V2G	Vehicle-to-grid.
Wh	Watthours.

I. INTRODUCTION

THE rapid deployment of EVs and renewable energy is primarily attributed to two critical factors: reducing dependency on fossil fuels and the increasing demand for low-carbon environmental solutions [1]. They also tackle various environmental concerns, such as air pollution. However, the fluctuating nature of essential renewable energy technologies like wind and solar could pose challenges to grid stability and supply security. To mitigate the impact of renewable energy uncertainty, V2G technology is proposed, which allows for bidirectional electricity flow between the grid and EVs, enabling the utilization of EVs as distributed portable energy storage systems integrated into the grid with controllable charging and discharging [2]. With coordinated V2G scheduling, EV batteries solely could satisfy short-term grid storage demand by as early as 2030 [3], [4].

When EVs can feedback energy, appropriate V2G scheduling can be leveraged to purchase the same amount of electricity at a reduced cost and feedback energy at higher prices to

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earn a profit. In the field of V2G charging scheduling, while extensive literature exists, this article narrows its focus to the centralized scheduling of V2G charging by profit-oriented operators. A significant thread in the V2G scheduling literature revolves around optimizing the charging for a predetermined group of EVs [5]. Most studies are built upon this framework, particularly applicable to residential V2G charging issues where distributed implementation is often crucial. Clement-Nyns et al. [6] proposed a distributed optimization algorithm for residential V2G charging that models power loss in the grid as a quadratic function and used dynamic programming to solve the quadratic programming problem. Gan et al. [7] proposed a distributed optimization framework to minimize the quadratic function of total load, demonstrating that a fixed number of EVs can converge to the optimal value under varying charging demands. Lee et al. [8] proposed an adaptive charging optimization algorithm based on convex optimization and MPC, accounting for three-phase imbalance and nonideal battery behavior, and validated its effectiveness through real-world deployment. While these methods can be adjusted to address the centralized V2G scheduling issue, their goals diverge significantly from those of commercial V2G charging operators. These models lack essential components such as the stochastic arrival of charging demands and the high cost of peak demand charges. In addition, despite the higher computational cost of centralized approaches, their solutions prove more profitable due to their capacity to optimize overall performance.

The complex and uncertain nature of large-scale V2G charges means that optimal operation is nontrivial. Thus, a suitable approach to address these challenges is economic MPC (economic MPC), which directly optimizes these cost functions in real time using a system model. Halvgaard et al. [9] presented one of the earliest implementations of economic MPC aimed at minimizing the charging costs for a determined set of EVs. Another study expands this framework to integrate EV charging with building energy management for co-optimization [10]. Risbeck and Rawlings [11] proposed an economic MPC with closed-loop optimization to solve time-varying prices and peak demand charges. These formulations lack consideration for the stochastic arrival of charging demand and the high costs associated with demand charges. Tang and Zhang [12] propose a low-complexity optimal MPC approach for EV charging scheduling without information about future EV arrivals. Yang et al. [13] presented a block MPC approach aiming to optimize the balance between maximizing profit in scheduling and minimizing penalties associated with peak demand charges during stochastic charging demand arrivals. By imposing a peak demand charge to promote coordination among stations and chargers, the potential of EVs as mobile energy storage can be fully realized. Shi et al. [14] proposed a distributed MPC algorithm of online peer-to-peer trading for EV charging strategies considering the unexpected departure of EV users. These papers considered the stochastic arrival of charging demand, the possibility of sudden departure by EV owners, and peak demand charge, effectively addressing the range anxiety and avoiding the cost of peak demand charges. However, some of the missing ingredients in these scheduling are V2G capability and battery degradation cost. Ebrahimi et al. [15] developed stochastic charging optimization of V2G-capable EVs with a comprehensive battery aging model. Razi et al. [16] proposed a predictive two-stage user-centered algorithm for V2G charging considering the state of health of the battery. For easy incorporation into online charging management, an MILP is introduced with a linear battery degradation model. Farzin et al. [17] investigated the major contributing factors to the degradation process of lithium-ion batteries in V2G applications, incorporating these factors into an MILP model to estimate battery life duration based on operational conditions. Cardoso et al. [18] applied a linear battery aging and degradation model to a multienergy microgrid sizing framework using MILP. The integration of this battery aging model into the broader microgrid sizing formulation is detailed. Khezri et al. [19] proposed a method for optimizing the V2G scheduling of an EV, incorporating a comprehensive battery degradation model that accounts for both calendar aging and cycle aging. However, these algorithms generally do not incorporate the common standard CC–CV charging mode for lithium batteries, leading to battery voltage exceeding the rated voltage during the later stages of charging, which accelerates battery aging [20]. Meanwhile, their approach overlooks the fluctuation in power rates. The absence of constraints on power change rates leads to significant fluctuations in charging and discharging power, potentially accelerating battery aging and equipment wear, thereby incurring additional costs.

Compared with existing charging scheduling methods, the contributions of this article are reflected in the following three aspects.

1) A CMPC-based V2G charging scheduling method is proposed. This method can avoid the penalty of peak demand charges and mitigate sharp fluctuations in V2G charging power.
2) An MILP optimization is established by considering range anxiety and battery degradation model, as well as integrating the CC–CV charging mode.
3) Instead of optimizing calculations for each time stage, a control package strategy and an online rolling optimization method are presented to reduce computational burden and consider stochastic charging demand arrivals.

II. Model Formulation

The scheduling of EV charging and discharging at a public facility under peak demand charge is investigated. EVs with charging needs arrive randomly, each with a specific energy request and deadline for completion.

A. Nominal Model Assumptions

Consider an EV charging station equipped with V2G capability and N dc chargers, as depicted in Fig. 1. The energy flow is bidirectional, allowing for two-way exchange between

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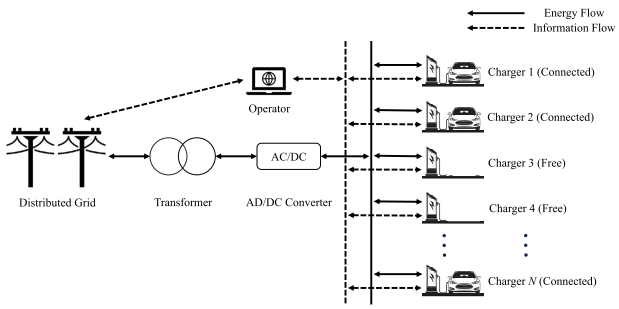

Fig. 1. Diagram of a charging station with V2G capability.

distributed power grids and chargers.[1] Charging occurs during periods of low electricity prices, while during high-price times, EVs feed power back to the grid, generating revenue via V2G. Information flow is also bidirectional. The operator receives electricity prices and power constraints from the distributed power grid and EV information from chargers. They then run charging and discharging optimization algorithms and send optimized control signals to chargers for optimal charging and discharging control. Simultaneously, the operator transmits overall charging and discharging power information back to the grid, facilitating optimization scheduling and power price adjustments in the distributed grid. Upon arrival, EVs are randomly connected to available chargers. Each EV reveals its charging requirements and completion deadline to the operator.[2]

Common states of lithium-ion batteries include the SOC, state of health, state of power, and SOE. Although SOC is widely used in the literature to represent battery charge, its measurement units vary between Ah and Wh, leading to inconsistencies. To address this ambiguity, advocacy for the use of the term SOE is promoted, which maintains clarity by focusing on energy units, as demonstrated by [20]. Further support for this preference is that energy market participants primarily transact in Wh rather than Ah.

A discrete scheduling time horizon is denoted by $\mathcal{T} = \{0, 1, \ldots, T\}$. To simplify the presentation, several assumptions regarding the V2G charging station are made as follows.

1) Each charger can only serve one EV at a time, and likewise, each EV can only be served by one charger at a time.
2) At the beginning of time interval t_o, EV arriving at the ith charger reveals stochastic energy completion states ($\mathrm{SOE}_{i,\mathrm{dep}}$) and departure times ($T_{i,\mathrm{dep}}$). Subsequently, when the deadline arrives, EVs will be automatically removed from the charging system.

[1]The proposed optimization algorithm can be applied to alternating current (ac) chargers as well, as the charging and discharging power for V2G is controlled independent of the type of charging current. However, since ac charging typically involves onboard chargers with power control integrated within the vehicle, effective implementation would require the use of IoT technology.

[2]In charging stations, customers cannot always wait until their EVs finish charging. Therefore, it is common practice to include a completion deadline in the charging demand. This allows customers to plan their return once charging is completed. However, achieving absolute precision in predicting charging deadlines by customers is challenging, and sudden departures may occur unexpectedly.

Remark 1: Assumption 1) is common practice for the majority of current charging stations [21]. Assumption 2) assumes that charging stations receive stochastic charging demand and departure time information from users when EVs arrive, which is crucial for optimizing charging algorithms for EVs. Some commercial charging stations may directly limit charging time [22].

B. V2G Scheduling Model

Throughout the day, EVs primarily remain idle, providing significant charging time and flexibility. Compared to traditional timed charging, V2G technology represents a more intricate approach, facilitating bidirectional energy exchange between EVs and the power grid. The scheduling of EV charging and discharging, whether through transfers or V2G operations, exhibits strong physical-time dependence. In this section, a mixed-integer linear programming (MILP) model for V2G scheduling is presented, subject to various constraints such as the CC–CV charging mode, maximum charging rates of batteries and chargers, dynamic electricity prices, and the maximum total charging power specified by the distribution grid.

1) External Random Inputs: The input to the V2G scheduling model consists of a stochastic vector process, simulating the random charging demands for individual chargers once connected to EV. The connection of each charger is modeled as a switch process; the charger remains occupied throughout the charging period of the EV and becomes idle upon the departure time. At the beginning time t_o of an occupancy duration for charger i, an EV arrives with stochastic charging demand $\mathrm{SOE}_{i,\mathrm{dep}}$ and stochastic departure time $T_{i,\mathrm{dep}}$. Hence, the input procedure at charger i is given by $\alpha_{i,t_o} = (\mathrm{SOE}_{i,\mathrm{dep}}, T_{i,\mathrm{dep}})$, where the scheduling horizon is $\mathcal{T} = \{t_o, \ldots, T_{i,\mathrm{dep}}\}$. When the charger is free, $\alpha_{i,t} = (0, 0)$.

2) Charger State Dynamics: The state of charger i at current time t is represented by $x_{i,t} = (\mathrm{SOE}_{i,t}, \beta_{i,t})$ [13]. Here, $\mathrm{SOE}_{i,t}$ denotes the SOE of the EV connected to charger i at current time t, and $\beta_{i,t} = T_{i,\mathrm{dep}} - t$ denotes the remaining service time. Thus, the state of charger i evolves according to the following formulation:

$$x_{i,t+1} = \begin{cases} x_{i,t} + \left(u_{i,t}, -\Delta t\right), & \text{if } \beta_{i,t} > \Delta t \\ \alpha_{i,t}, & \text{if } \beta_{i,t} \leq \Delta t \end{cases} \tag{1}$$

where $u_{i,t} = (P_{i,t}/E_{i,\mathrm{cap}})$. $E_{i,\mathrm{cap}}$ represents the battery capacity of the EV connected to charger i, $P_{i,t}$ denotes the charging or discharging power at current time t, and Δt represents the time interval from t to $t+1$. Note that if no EV connects to charger i, the status of charger i stays constant at $(0, 0)$.

3) CC–CV Charging Mode: The most appropriate charging mode for lithium-ion batteries is considered to be the CC–CV mode [23]. Initially, the battery undergoes charging with a CC, causing the voltage of the battery cell to rise. For the CC mode, a constant maximum charging power $P_{i,\mathrm{cha}}$ is assumed. When the voltage reaches its maximum value $V_{\max}$ at a specific $\mathrm{SOE}_i^{\text{cc-cv}}$, the charging method is altered to prevent battery degradation. At this point, the voltage is maintained constant (CV) while the current is exponentially reduced. This

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behavior is approximated by a piecewise linear simplification of the actual data [24]. The model profile can be expressed mathematically as

$$P_{i,t}^{\max} = \begin{cases} P_{i,\text{cha}}, & \text{if } \text{SOE}_{i,t} \le \text{SOE}_i^{\text{cc-cv}} \\ \dfrac{P_{i,\text{cha}} \cdot (1 - \text{SOE}_{i,t})}{1 - SOE_i^{\text{cc-cv}}}, & \text{if } \text{SOE}_{i,t} > \text{SOE}_i^{\text{cc-cv}} \end{cases} \tag{2}$$

where $P_{i,t}^{\max}$ is the maximum charger power of the ith EV. The $\text{SOE}_i^{\text{cc-cv}}$ for transitioning from the CC to the CV mode typically varies based on the charging power level and the specific battery charging management system. The $\text{SOE}_i^{\text{cc-cv}}$ is assumed to be 85% [20].

4) Range Anxiety: EV users face two primary concerns regarding V2G technology: range anxiety and battery degradation. In this article, range anxiety refers to the uncertainty of sudden departure. Concerns arise that V2G operations might drain the EV battery to a level insufficient for immediate travel. Due to uncertainties stemming from users' inaccurate predictions of departure times and the possibility of retrieving their vehicles at any moment, the proposed scheduling method aims to not only complete EV charging tasks within their deadlines but also ensure that the battery maintains a range sufficient to meet daily needs, particularly during V2G operations. In this article, to alleviate users' range anxiety, the SOE of each EV is maintained above SOE threshold SOE_i^{th} after time threshold t_i^{th} hours of charging, ensuring the battery meets uncertainty driving needs. That can be expressed mathematically as

$$\begin{cases} \text{SOE}_{i,t} = \text{SOE}_{i,t_o}, & \text{if } t = t_o \\ 0 \le \text{SOE}_{i,t} \le 1, & \text{if } t_o \le t \le t_o + t_i^{\text{th}} \\ \text{SOE}_i^{\text{th}} \le \text{SOE}_{i,t} \le 1, & \text{if } t_o + t_i^{\text{th}} \le t \le T_{i,\text{dep}} \\ \text{SOE}_{i,t} \ge \text{SOE}_{i,\text{dep}}, & \text{if } t \ge T_{i,\text{dep}}. \end{cases} \tag{3}$$

Note that this threshold can be easily extended to any SOE and time level predicted by statistical data. In this article, it is assumed that an SOE threshold of $\text{SOE}_i^{\text{th}} = 40\%$ is sufficient to meet users' daily needs,[3] and achieving this level after a time threshold of $t_i^{\text{th}} = 4$ h can alleviate users' range anxiety. Moreover, the MPC rolling optimization framework is employed to mitigate the uncertainty of EV owners' charging behavior on V2G operations. In the event of an unexpected vehicle departure, a new optimization cycle is initiated, excluding already departed vehicles, thereby reducing the uncertainty in charging behavior's impact on V2G operations.

5) Battery Degradation Model: To mitigate battery degradation anxiety, a linear battery degradation model for V2G is incorporated into the designed objective function, considering the detailed battery degradation models for calendar aging and cycle aging proposed in [18] and [19].[4] This helps reduce the frequency of EV charging and discharging and extends the battery lifespan.

The predetermined coefficients for both calendar and cycling aging in the battery degradation model are derived from a substantial amount of experimental data, specifically from the empirical aging model developed and experimentally validated by Wang et al. [25]. A semi-empirical model was introduced by incorporating key experimental parameters such as temperature, time, depth of discharge, and charge/discharge rate. This model effectively decouples the capacity losses caused by cycling and calendar aging, allowing for their individual contributions to be summed, thereby providing an accurate representation of the total capacity loss over any given time horizon. The degradation model of the EV's battery at each time interval is the sum of the calendar aging and the cycle aging and is given by

$$D_{i,t}^{\text{tot}} = D_{i,t}^{\text{cal}} + D_{i,t}^{\text{cyc}} \tag{4}$$

where $D_{i,t}^{\text{cal}}$ and $D_{i,t}^{\text{cyc}}$ represent the percentage of capacity loss induced by calendar aging and cycle aging, respectively.

Calendar aging, as derived by [18], is strongly affected by temperature (K) and time (t) and can be expressed as

$$D_{i,t}^{\text{cal}} = \theta \epsilon^{\frac{E_a}{RK}} \sqrt{t} \tag{5}$$

where θ, ϵ, E_a, and R are predetermined calendar aging coefficients. K is the absolute temperature in Kelvins. t is the total duration in days.

The cycle aging model of the battery [25] is given by

$$D_{i,t}^{\text{cyc}} = (\alpha K^2 + \beta K + \gamma)\epsilon^{(\sigma K + \epsilon) I_{i,t}^{\text{rate}}} \text{Ah}_{i,t} \tag{6}$$

where α, β, γ, ϵ, and σ are predetermined cycle aging coefficients.

$I_{i,t}^{\text{rate}}$ is the charge/discharge rate at time t expressed as a C-rate [19]

$$I_{i,t}^{\text{rate}} = \frac{\delta_{i,t,\text{cha}} \cdot P_{i,t,\text{cha}} + \delta_{i,t,\text{dis}} \cdot P_{i,t,\text{dis}}}{E_{i,\text{cap}}} \tag{7}$$

where $\delta_{i,t,\text{cha}}$ and $\delta_{i,t,\text{dis}}$ are Boolean variables. Because the battery cannot charge and discharge simultaneously, the Boolean variable $\delta_{i,t,\text{cha}}$ is set to 1 when charger i is charging and 0 otherwise, while the Boolean variable $\delta_{i,t,\text{dis}}$ is set to 1 when charger i is discharging and 0 otherwise. This prevents simultaneous charging and discharging of the battery. $P_{i,t,\text{cha}}$ and $P_{i,t,\text{dis}}$ represent the battery charging and discharging power, respectively.

$\text{Ah}_{i,t}$ is the Ah-throughput, which can be defined as $\text{Ah}_{i,t}$ = (depth of discharge) * (battery capacity) * (cycle number). Since the relationship between the depth of discharge and cycle number requires empirical data from battery manufacturers that are typically difficult to obtain, the cumulative charge has been used as an alternative for $\text{Ah}_{i,t}$

$$\text{Ah}_{i,t} = \frac{\delta_{i,t,\text{cha}} \cdot P_{i,t,\text{cha}} + \delta_{i,t,\text{dis}} \cdot P_{i,t,\text{dis}}}{V} \Delta t \tag{8}$$

where V is the nominal voltage of the battery, assumed to remain constant during the battery's operation.

[3]The average range of EVs currently reaches around 500 km, so an SOE of 40% corresponds to approximately 200 km of driving range. This is sufficient to meet most short-term driving needs. Setting the SOE threshold at 40% strikes a balance between usability and energy flexibility. As EV ranges continue to improve, this threshold can be gradually lowered, allowing for greater adjustable energy potential in V2G applications.

[4]This article primarily focuses on designing the CMPC algorithm for the V2G scheduling problem. Consequently, the battery degradation model utilized is directly based on the models proposed in [18] and [19].

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By summing (5) and (6), the general function for the total capacity loss of the battery is

$$D_{i,t}^{\text{tot}} = (\alpha K^2 + \beta K + \gamma)\epsilon^{(\sigma K+\epsilon)I_{i,t}^{\text{rate}}} A_h + \theta\epsilon^{\frac{E_a}{RK}}\sqrt{t}. \quad (9)$$

This equation describes the loss of electric storage capacity by considering four crucial experimental parameters: temperature, time, depth of discharge, and charge/discharge rate. The implementation of this equation in an MILP formulation can be achieved by linearizing it within specific domains [18].

Remark 2: The total battery degradation, which is influenced by cumulative discharge, temperature, time, and charging current, is typically modeled by (9). This model suggests that battery degradation exhibits an exponential relationship with temperature and C-rate. However, under specific operating conditions, such as a charging C-rate below 0.5 and temperatures maintained between 15 °C and 30 °C (288–303 K), the partial derivatives of capacity loss with respect to current and temperature, $(\partial D_t^{\text{tot}}/\partial I_t^{\text{rate}})$ and $(\partial D_t^{\text{tot}}/\partial K)$, can be approximated as zero [18]. This approximation allows the capacity loss to be treated as a linear function of time and cumulative output, greatly simplifying the analysis and application of predictive models within these standard operating ranges.

The degradation cost of V2G charging can be expressed as

$$C_{i,t,\text{deg}} = (\text{BP} - \text{SP}) \cdot D_{i,t}^{\text{tot}} / (1 - E_n) \quad (10)$$

where BP and SP are the battery price and scrap price of the battery, respectively. E_n is the end of battery life. In (10), the battery is considered to have reached the end of its life when its capacity drops below 80% of its initial capacity [15].

6) Physical Constraints: The constraints on EV charging and discharging power can be categorized into two aspects: physical hard constraints and peak demand charge. Physical hard constraints encompass the maximum charging power tolerated by EV batteries, the maximum power delivery capacity of chargers, and the inability of batteries to charge and discharge simultaneously. The physical hard constraints can be expressed mathematically as

$$P_{i,t} = \delta_{i,t,\text{cha}} \cdot P_{i,t,\text{cha}} + \delta_{i,t,\text{dis}} \cdot P_{i,t,\text{dis}} \quad (11)$$

$$0 \le P_{i,t,\text{cha}} \le \min\left\{P_{i,\text{cha}}^{\text{bat}},\ P_{i,\text{cha}}^{\text{sta}}\right\} \quad (12)$$

$$\max\left\{P_{i,\text{dis}}^{\text{bat}}, P_{i,\text{dis}}^{\text{sat}}\right\} \le P_{i,t,\text{dis}} \le 0 \quad (13)$$

$$0 \le \delta_{i,t,\text{cha}} + \delta_{i,t,\text{dis}} \le 1 \quad (14)$$

where $P_{i,\text{cha}}^{\text{bat}}$ and $P_{i,\text{cha}}^{\text{sta}}$ denote the maximum charging power supported by the battery and the charging stations, while $P_{i,\text{dis}}^{\text{bat}}$ and $P_{i,\text{dis}}^{\text{sat}}$ indicate the maximum discharging power supported by the battery and the charging stations, respectively. Equation (14) prevents simultaneous charging and discharging of the battery by restricting the sum of the Boolean variables $\delta_{i,t,\text{cha}}$ and $\delta_{i,t,\text{dis}}$.

7) Peak Demand Charge: The distributed grid imposes a soft constraint on peak demand at charging stations within the V2G system by introducing peak demand charges, thereby fostering collaboration among the stations. The implementation of peak demand charge plays a crucial role in optimizing the interaction between the grid and EVs. For the grid operator, peak demand charge mitigates the impact of peak loads by encouraging coordinated charging behaviors among charging stations, allowing for efficient management of charging loads, preventing simultaneous charging or discharging across multiple EVs, and ultimately enhancing grid stability. This also allows the grid to better harness the potential of V2G technology, increasing overall grid capacity and reliability. For EV owners, the peak demand charge presents an opportunity for increased V2G earnings. As the grid efficiently manages charging resources and expands its capacity, the cost savings from optimized dispatching can be shared with EV owners, offering them greater financial incentives for participating in V2G programs. This dual benefit underscores the significance of peak demand charge in fostering a sustainable and profitable V2G ecosystem [11].

This charge, defined by the distributed grid, sets maximum charging and discharging power limits for charging stations within each time interval Δt. Violating these limits incurs extra charges. These peak demand charges can be expressed as

$$\mathcal{V}_t = \eta \cdot \min\left\{\left|\sum_{i=1}^{N} P_{i,t}\right| - M, 0\right\} \quad (15)$$

where η signifies penalties for exceeding peak demand charges and N denotes the total number of chargers owned by the charging station.

M, which represents the peak demand charge at each interval Δt, is given as follows:

$$M = \frac{1}{3} \cdot \sum_{i=1}^{N} P_{i,t,\text{cha}}. \quad (16)$$

The excess charging and discharging energy from the charging station beyond the peak demand charging incur a cost of η. The peak demand charge M is typically established through advanced load forecasting techniques at the distribution level, often supplied by the distributed grid itself [26]. For coordination, the peak demand charge is simplified by setting it to one-third of the peak charging and discharging capacity at each interval.

8) Control Law Maps: V2G scheduling aims to determine the optimal control policy $u_{i,t}^{\star}$ that maximizes the expected total reward at the charging station. This can be represented by a control law mapping $\mathbf{u}_{i,t}$ to controls at each interval Δt

$$\mathbf{u}_{i,t}(x_{i,t}, \alpha_{i,t}) = \left(u_{i,t}^{\star}, u_{i,t+1}^{\star}, \ldots, u_{i,t+W-1}^{\star}\right) \quad (17)$$

where W refers to the predicted duration of time and $(u_{i,t}^{\star}, u_{i,t+1}^{\star}, \ldots, u_{i,t+W-1}^{\star})$ are the optimal solutions to the presented scheduling optimization problem. The control law mapping is a Markov decision process that depends only on the current state and predicted states, making it naturally suitable for predictive control-based optimization scheduling [13].

III. Coordinated Predictive Control-Based Scheduling Algorithm

A CMPC-based scheduling approach is proposed to tackle four key challenges in V2G charging scheduling.

1) *Peak Demand Charge and Power Fluctuation:* The CMPC optimization incorporates peak demand charges and a penalty term for the rate of power change into

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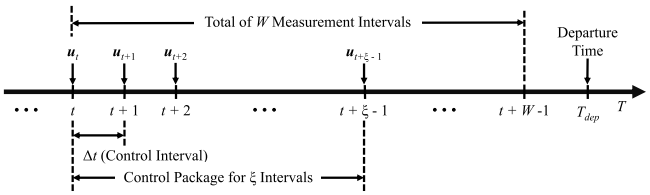

Fig. 2. Temporal structure of the CMPC algorithm.

the cost function to coordinate among chargers, thereby avoiding high peak demand costs and mitigating battery and charging equipment aging.

2) *Battery Degradation:* In CMPC optimization, a detailed battery degradation model for both calendar aging and cycle aging is considered, along with the standard CC–CV charging mode, to prevent frequent charging and discharging cycles and ensure that the battery voltage does not exceed the rated voltage in the final charging stage. This approach aims to extend battery life and alleviate concerns about battery aging.

3) *Range Anxiety:* Range anxiety involves the uncertainty of sudden departures, along with concerns that V2G might drain the EV battery to the point where it cannot fulfill the user's immediate departure requirements. By setting time and SOE thresholds, the proposed scheduling method not only completes EV charging tasks within deadlines but also ensures that the battery maintains sufficient range during V2G operations to meet the driving needs of unexpected departures.

4) *Computational Burden and Stochastic Arrivals Demand:* Traditional MPC involves making a single decision per interval, which can be computationally intensive. In contrast, inspired by [27], [28], and [29], a control packet strategy is proposed. This strategy is typically used in networked control systems to address delay and packet loss issues. Applying it to the proposed CMPC algorithm helps reduce the computational load and communication pressure. As shown in Fig. 2, during the CMPC rolling optimization process, each computation packages the optimized control signals, including all optimized control signals from time t to $t+\xi-1$ ($\xi \leq W$). Reoptimization occurs only when a new vehicle arrives or when reaching time $t+\xi-1$. By utilizing the control packet strategy and rolling optimization techniques, the stochastic charging demand arrivals can be effectively managed, and the computational and communication pressures can be reduced.

To this end, the CMPC-based scheduling problem solved by the EV charging station using the MILP framework is given as follows (18). MILP models are crucial for V2G scheduling as integer variables allow EVs to charge or discharge exclusively, while linear programming ensures quicker solving speed compared to nonlinear methods, meeting the demand for real-time scheduling

$$\min\ J(t) = \sum_{k=t}^{t+W-1} \mathcal{Y}_k + \Delta\mathcal{U}_k + \mathcal{V}_k$$
$$\text{s.t. } (1)\text{–}(3), (10)\text{–}(15) \tag{18}$$

Algorithm 1 CMPC Algorithm

Initialization: Initialize system with x_{i,t_o}, α_{i,t_o}, t_i^{th}, SOE_i^{th}, M, ξ, W.
for $t \in \{0, 1, 2, \ldots, T\}$ **do**
  Observe the system current state $x_{i,t}$ and $\alpha_{i,t}$.
  Update constraints (1)(2)(3)(11)(12)(13)(14).
  if $\mathcal{L} \geq \xi$ **or** EV arrivals **then**
    Solve the CMPC Problem (18).
    $\mathbf{u}_{i,t}(x_{i,t}, \alpha_{i,t}) = \text{CMPC}(x_{i,t}, \alpha_{i,t}, J(t))$.
    Extract the first package (ξ intervals) of controls from the solution (18): $\{u^\star_{i,t}, \ldots, u^\star_{i,t+\xi-1}\}$.
    Refresh the predictive control counter $\mathcal{L}$: $\mathcal{L} = 1$.
  end if
  for $i \in \{0, 1, 2, \ldots, N\}$ **do**
    Take $u^\star_{i,t}$ as the decision at t,
    Update the system state to $x_{i,t+1}$ according to (1).
  end for
  Update the predictive control counter $\mathcal{L}$: $\mathcal{L} = \mathcal{L} + 1$.
end for

where

$$\mathcal{Y}_t = \sum_{i=1}^{N} C_{i,t,\text{cha}} + C_{i,t,\text{dis}} + C_{i,t,\text{deg}}$$
$$C_{i,t,\text{cha}} = \eta_{i,\text{cha}} \cdot \lambda_t \cdot P_{i,t,\text{cha}} \cdot \Delta t$$
$$C_{i,t,\text{dis}} = \eta_{i,\text{dis}} \cdot \lambda_t \cdot P_{i,t,\text{dis}} \cdot \Delta t$$
$$\Delta\mathcal{U}_t = \sum_{i=1}^{N} K_i \cdot \left|\frac{du_{i,t}}{dt}\right| = \sum_{i=1}^{N} \frac{K_i}{E_{i,\text{cap}}} \left|\frac{dP_{i,t}}{dt}\right|$$

where variables $\eta_{i,\text{cha}}$ and $\eta_{i,\text{dis}}$ denote the charging and discharging efficiency of chargers, respectively. In V2G scheduling, it is typically assumed that the cost of power received and injected into the grid is equal, which is unrealistic. Hence, a coefficient $\eta_{i,\text{dis}}$ is introduced in this work to account for factors like revenue for the grid owner or power losses. In this article, $\eta_{i,\text{cha}}$ is set to 1 and $\eta_{i,\text{dis}}$ is set to 0.9. In addition, λ_t represents dynamic electricity prices. K_i is the weight of manually set smoothing charging or discharging power. In practical applications, K_i can be adjusted according to the system's sensitivity to power fluctuations. If minimizing the frequency and magnitude of energy fluctuations is a priority, K_i should be set higher; otherwise, it can be lower.

Remark 3: By combining (11) and (18), it can be seen that in the proposed MILP model, the decision variables are $\delta_{i,t,\text{cha}}$, $\delta_{i,t,\text{dis}}$, $P_{i,t,\text{cha}}$, and $P_{i,t,\text{dis}}$. The variables $\delta_{i,t,\text{cha}}$ and $\delta_{i,t,\text{dis}}$ are integers and Boolean values, ensuring that the battery cannot charge and discharge simultaneously, and allowing for the adaptation to different power losses associated with charging and discharging. The variables $P_{i,t,\text{cha}}$ and $P_{i,t,\text{dis}}$ represent the magnitudes of charging and discharging power, respectively, and are used to calculate the charging cost, battery aging cost, and discharging revenue.

The CMPC-based scheduling approach is summarized as Algorithm 1. The cost function $J(t)$ of the CMPC-based scheduling controller is composed of three elements. The first

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part $\mathcal{Y}_t$ reflects the charging and discharging cost within the prediction window $t + W$, encompassing charging electricity costs, discharging revenue, and battery aging costs associated with V2G. The second part $\Delta\mathcal{U}_t$ accounts for the change rate of charging and discharging power, aiming to mitigate sharp fluctuations in power. The third part $\mathcal{V}_t$ ensures adherence to a soft system constraint, facilitating coordinated charging and discharging activities across the network to prevent the high cost of peak demand charge. By minimizing $J(t)$, the obtained strategies not only optimize energy costs associated with grid interactions but also alleviate user concerns regarding range and battery longevity in V2G.

Remark 4: Unlike traditional MPC methods [30], [31] in process control engineering, which aim to minimize the difference between the output of a controlled system and a reference value, CMPC prioritizes maximizing the total reward for EV charging applications, without considering stability and tracking.

Remark 5: MPC rarely achieves a global optimal solution. In this study, the arrival demand of EVs is assumed to be entirely random and unpredictable. Consequently, the presented CMPC algorithm only ensures a local optimal solution based on the current known vehicle demand, rather than a global optimal with complete information. This makes the proposed algorithm more realistic, especially for small charging stations where customer behavior is unpredictable.

Remark 6: Algorithm 1 can be directly solved using commercial solvers like Gurobi and CPLEX, which are more efficient compared to other intelligent algorithms such as evolutionary algorithms and particle swarm optimization algorithms [32]. Due to the integer nature of decision variables in (18), the complexity of CMPC exponentially increases with the number of pending tasks. However, modern commercial solvers efficiently resolve MILPs with thousands of variables and the computational duration of the proposed method is within an acceptable range [33].

IV. Simulation Results

The proposed CMPC online algorithm, detailed in Section III, is evaluated through three charging scenarios: absence of both smoothness and coordination, lack of coordination alone, and incorporation of both smoothness and coordination. Table I presents the arrival and expected departure times, initial SOE, targeted final SOE, and the specifications for various EVs [16]. The parameters of the EVs are presented in Table II. It is presumed that the time-varying electricity prices were deterministic and sourced from LCG Consulting's online energy price database.[5] These simulations are executed on a computer equipped with an i7-13700F GPU and 16 GB RAM, utilizing MATLAB, YALMIP, and Gurobi for computational processes.

A. Absence of Both Smoothness and Coordination Case

In this case, there is no smoothness and coordination between the four EVs, which means that the smoothness and coordination terms in the cost function (18) are zero, that is, $\Delta\mathcal{U}t = 0$ and $\mathcal{V}t = 0$. The scheduling results of the four individual chargers without smoothness and coordination are shown in Fig. 3 and Table III. The simulation clearly shows that EVs can optimize charging during low-price periods, discharge during high-price periods for V2G revenue, and simultaneously maintain the desired SOE levels. However, due to the lack of coordination, each EV only considers maximizing its own profit, leading to the total charging and discharging power exceeding the peak demand charges. This results in additional expenses for the charging station, which are ultimately factored into the overall cost. Due to the lack of a smoothness penalty term, there are significant fluctuations in the charging and discharging power, which bring additional battery and charging equipment aging costs. In addition, the absence of both smoothness and coordination leads to frequent charging and discharging, increasing battery aging costs. Including the peak demand charge and the additional battery aging costs in the final cost results in the highest charging cost of \$134.15.

TABLE I

Initial State From EVs

	EV1	EV2	EV3	EV4
t_o	03:00	06:00	10:00	10:00
t_{dep}	21:00	23:00	22:00	23:30
SOE_o	20%	40%	50%	80%
SOE_{dep}	80%	100%	90%	40%
E_{cap}	27.33 kWh	35.74 kWh	50.44 kWh	36.09 kWh
$P_{min/max}$	-15.5 / 15.5 kW	-18 / 18 kW	-23.54 / 23.54 kW	-18.5 / 18.5 kW

TABLE II

Detailed Parameters for EVs and Degradation Models

Parameter	Value
SOE_i^{th}, t_i^{th}, Δt, ξ, W	0.4, 4 h, 5 min, 2 h, 10 h
$P_{i,cha}^{sta}$, $P_{i,dis}^{sta}$, K_i	26.32 kW, -26.32 kW, 5
θ, ϵ, E_a, R, K	4944 months$^{-1/2}$, 2.35 h, 24500 Jmol^{-1}, 8.314 Jmol^{-1}K^{-1}, 298 K
α, β, γ, σ, V	5.04e-6 kW$-$1K^{-2}, -2.998e-3 kW^{-1}K^{-1}, 0.446kW^{-1}, 3.85 V
BP, SP, η, M	\$7500, \$2500, \$20/kWh, 2.09 kWh

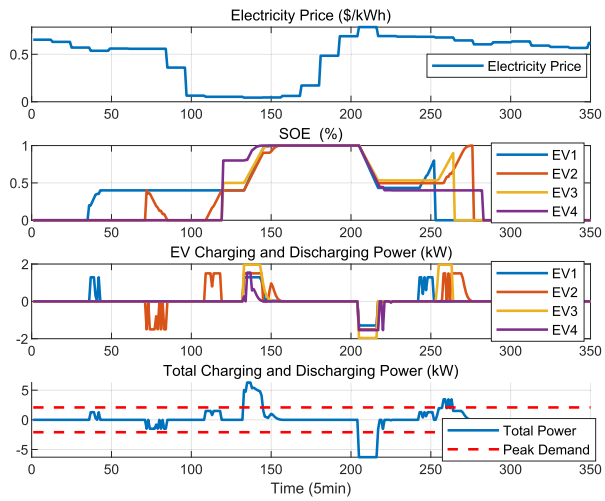

Fig. 3. Scheduling results of the four individual chargers without both smoothness and coordination.

B. Lack of the Smoothness Case

In this case, there is a lack of smoothness for the four EVs, which means that the smoothness terms in the cost

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TABLE III
COST COMPARISON OF FOUR EVS FOR DIFFERENT CASES

	EV1					EV2					EV3					EV4					Total Cost
	Charge Cost	Disharge Reward	Aging Cost	Peak Demand Charge	Total Cost	Charge Cost	Disharge Reward	Aging Cost	Peak Demand Charge	Total Cost	Charge Cost	Disharge Reward	Aging Cost	Peak Demand Charge	Total Cost	Charge Cost	Disharge Reward	Aging Cost	Peak Demand Charge	Total Cost	
Without Smoothess and Coordination	10.51	-10.96	0.36	32.50	32.41	13.55	-19.91	0.66	41.86	36.16	13.62	-16.65	0.51	54.33	51.82	0.31	-15.05	0.22	29.27	13.75	134.15
Lack of Smoothness	3.71	-3.86	**0.20**	**0**	**0.06**	1.74	-7.17	**0.38**	**0**	**-5.04**	1.19	-3.56	**0.23**	**0**	**-2.14**	0.33	-14.72	**0.22**	**0**	**-14.17**	**-21.30**
Lack of Coordination	10.52	-10.96	0.36	31.63	31.55	13.55	-19.91	0.66	37.72	32.02	13.62	-16.65	0.51	49.50	46.99	0.31	-15.05	0.22	27.37	12.85	123.43
With Smoothess and Coordination	3.68	-3.63	**0.20**	**0**	**0.25**	1.77	-7.17	**0.38**	**0**	**-5.02**	1.20	-3.56	**0.23**	**0**	**-2.13**	0.34	-14.96	**0.22**	**0**	**-14.39**	**-21.30**

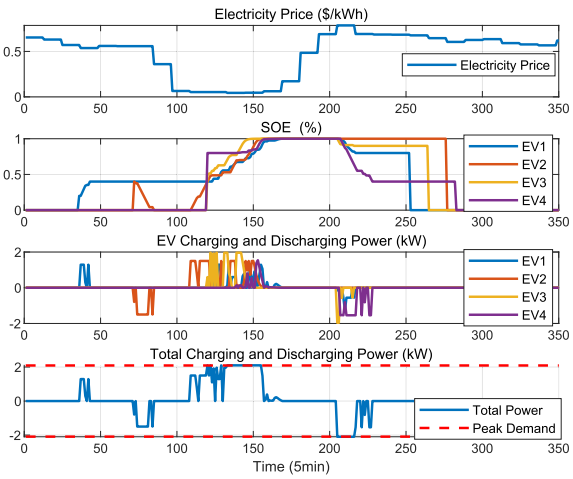

Fig. 4. Scheduling results of the four individual chargers without smoothness.

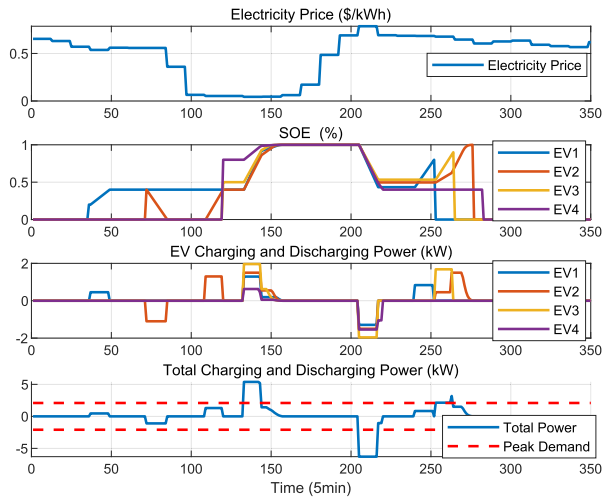

Fig. 5. Scheduling results of the four individual chargers without coordination.

function (18) are zero, that is, $\Delta\mathcal{U}_t = 0$. Due to the inclusion of coordination terms, the charging and discharging power of the batteries will not exceed the peak demand, thus avoiding high peak demand charges. However, as shown in Fig. 4 and Table III, the absence of smoothness constraints leads to significant fluctuations in the charging and discharging power. This will potentially reduce the lifespan of the batteries and the charging/discharging equipment.[6] However, since the potential aging cost due to power fluctuations is difficult to quantify, it has not been included in the overall cost calculation in this article. Therefore, even though the lack of smoothness case can achieve a relatively good total cost \$−21.30, it cannot be considered the optimal charging and discharging strategy due to the potential costs brought by significant fluctuations. The strategy that ignores the smoothness of V2G charging power cannot be regarded as the optimal V2G strategy, which is a point overlooked by most existing papers.

C. Lack of the Coordination Case

In this case, there is a lack of coordination between the four EVs, which implies that the coordination terms in the cost function (18) are zeros, that is, $\mathcal{V}_t = 0$. The scheduling results of the four individual chargers without coordination are shown in Fig. 5 and Table III. Due to the inclusion of a smoothness penalty term, the rate of change in charging and discharging power is reduced, resulting in smoother charging and discharging curves and a decrease in the peak power of charging and discharging. However, as depicted in Fig. 5 and Table III, due to the absence of a coordination term, the total charging and discharging power still exceeds the peak demand charges. This will lead to additional expenses for the charging station, which will ultimately be calculated into the overall cost \$123.43. In addition, the lack of coordination terms will also result in frequent charging and discharging of the batteries, increasing the battery aging costs.

D. Smoothness and the Coordination Case

In this case, the results of the four individual chargers using the proposed scheduling method, which includes both smoothness and coordination, are shown in Fig. 6 and Table III. By adding a coordination term to the cost function (18), the chargers can coordinate charging while meeting the EVs' charging demands, thus avoiding peak demand charges. In addition, by incorporating a smoothness term into the cost function (18), the chargers can avoid frequent fluctuations in charging and discharging power during coordinated charging, reducing battery and charging equipment aging. Through the combined effect of the smoothness term $\mathcal{V}t$ and the coordination term $\Delta\mathcal{U}t$, the EVs' charging and discharging power remain within the peak demand limits, ensuring smooth charging and discharging control and reducing overall costs. This allows V2G scheduling to not only offset charging costs

[6]The aging cost model used in the proposed algorithm is based on the models from [19] and [18]. However, these models do not account for the impact of power fluctuations on battery aging. Although studies such as [34] have highlighted the negative effects of power fluctuations on battery life, a quantifiable method to incorporate this impact into the aging model has not yet been developed.

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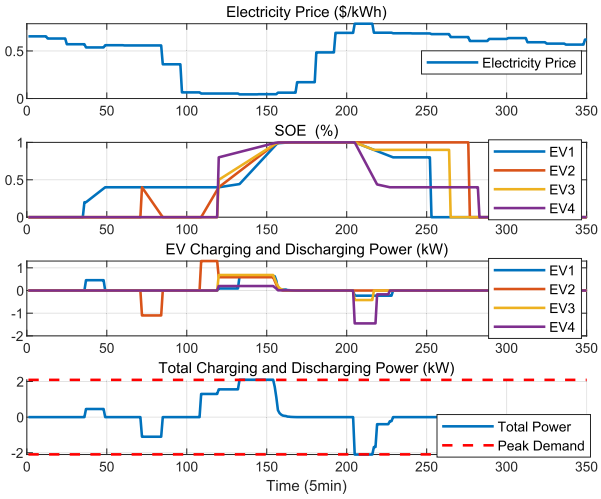

Fig. 6. Scheduling results of the four individual chargers with both smoothness and coordination.

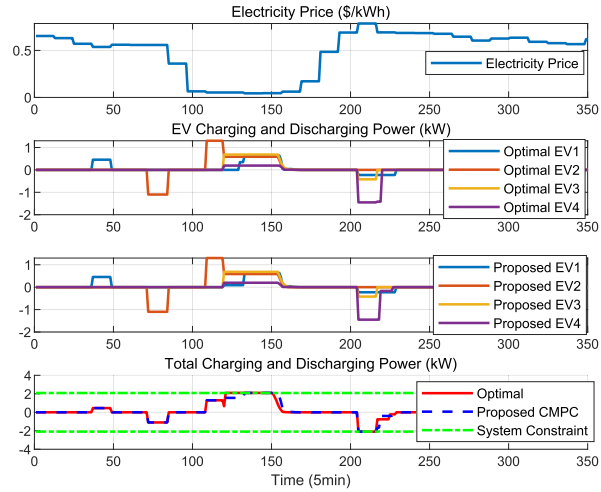

Fig. 7. Comparison of optimal scheduling results.

and battery aging expenses but also generate positive revenue. Consequently, the total cost amounts to $−21.30. Compared to the lack of coordination case, the proposed scheduling method avoids high peak demand charges. Compared to the lack of smoothness case, it avoids frequent fluctuations in charging and discharging power, potentially reducing equipment wear and battery aging during the V2G process.

E. Comparison With Optimal and Single-Step MPC Cases

In this case, the proposed method is compared with optimal scheduling and single-step MPC approaches, as depicted in Fig. 7 and Table IV. Optimal scheduling refers to the best solution obtained using the same cost function, where all vehicle and price information is known without any uncertainties. It can be observed that both optimal scheduling and the proposed CMPC scheduling have a total cost of $−21.30, and the power charts for charging and discharging are very close, demonstrating the superiority of the proposed method. The single-step MPC approach involves predictive optimization at each interval, using the first optimal solution to control the charging and discharging of EVs, and differs from the proposed CMPC algorithm by not utilizing control package technology. When optimization is performed at each step, the total runtime of the optimization program is 1848 s. In contrast, the runtime for the proposed CMPC algorithm, utilizing control package technology, is only 31.83 s. This demonstrates that the use of control package technology effectively reduces the number of optimization calculations, thereby alleviating computational load and communication pressure. Furthermore, comparisons with optimal scheduling indicate that the proposed algorithm effectively ensures overall system performance.

F. Comparison With Existing Methodologies

To demonstrate the effectiveness of our proposed CMPC algorithm, following methods are compared.

1) *Rule-Based Method:* V2G charging operates without coordination, focusing solely on electricity prices and charging demand, with user needs considered near the charging deadline to meet the target [35].
2) *Day-Ahead Method:* V2G charging requires precise prediction of the arrival time of EVs one day in advance, followed by charging and discharging planning based on vehicle demand and electricity prices [36].
3) *Economic MPC Method:* V2G charging operates in real time, updating the optimization strategy based on charging demand and electricity prices, effectively handling the randomness of EVs [11].

TABLE IV

COST COMPARISON OF OPTIMAL AND THE PROPOSED CMPC SCHEDULING

	Charge Cost	Discharge Reward	Aging Cost	Peak Demand Charge	Total Cost
Optimal	6.99	-21.30	1.04	0	-21.30
Proposed CMPC	6.99	-21.30	1.04	0	-21.30

TABLE V

COST COMPARISON OF METHODOLOGICAL APPROACHES

	Charge Cost	Discharge Reward	Aging Cost	Peak Demand Charge	Total Cost	Computing Time
Rule-based	74.79	-77.21	2.17	180.69	334.88	0.62
Day-ahead	8.84	-23.20	0.85	0	-13.49	9.77
Ecnomic MPC	8.84	-23.20	0.85	0	-13.49	517.25
Proposed CMPC	6.99	-29.34	1.04	0	**-21.30**	31.83

As shown in Fig. 8 and Table V, the rule-based method quickly optimizes calculations but incurs a high cost of peak demand charge due to the lack of coordination with other EVs. Consequently, the rule-based method requires only 0.62 s of computation time but results in a total cost of $334.88. The day-ahead method achieves relatively good optimization results and computation time, requiring only a single optimization operation, 9.77 s, and allowing for coordination. However, it requires precise prediction of vehicle arrival times, making it difficult to apply in practice. In addition, it does not account for the impact of the CC–CV charging mode. To ensure that battery voltage does not exceed the rated level, the SOE range needs to be limited to below 90%, which hinders the full potential of V2G, resulting in a total cost of only $−13.49. The economic MPC method addresses the uncertainty of vehicle

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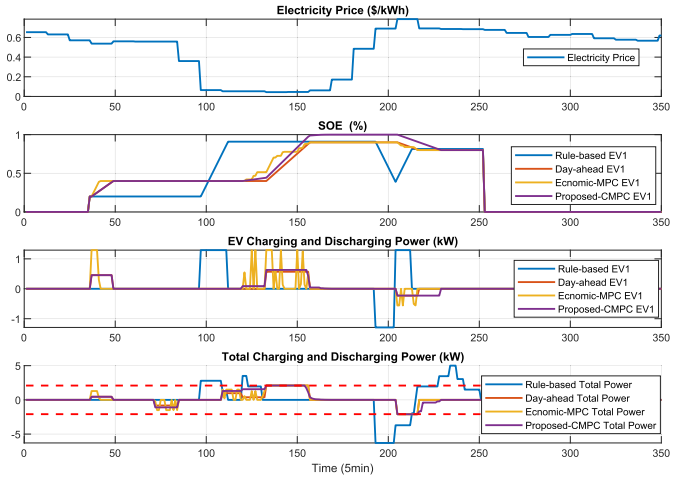

Fig. 8. Comparison of optimal scheduling results.

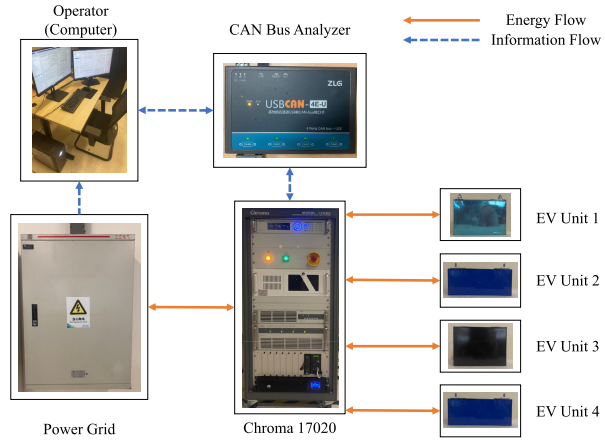

Fig. 9. Experimental platform of V2G scheduling optimization.

arrivals through real-time rolling optimization, but it also fails to consider the CC–CV charging mode and power fluctuations, leading to an inability to fully realize V2G potential. Like the day-ahead method, it results in a total cost of $−13.49 and experiences fluctuations in charging and discharging power. Furthermore, due to the lack of control package technology, the computational burden is high, with the total computation time for the entire day's optimization cycle being 517.25 s. The proposed CMPC method, by accounting for the CC–CV charging mode, fully harnesses V2G potential, achieving a total cost of $−21.30. It also effectively addresses the uncertainty of vehicle arrivals through real-time rolling optimization, and by utilizing control package technology, it significantly reduces computation time, requiring only 31.83 s for the entire day's optimization.

G. Scalability of the Proposed Algorithm

The proposed method is scalable, as demonstrated by simulations with 20, 50, and 100 EVs where arrival times and charging demands were randomly generated. The maximum time taken for a single optimization run was 10.12, 18.05, and 42.16 s, respectively, indicating the scalability of our algorithm. The total computation time for the entire day's optimization cycle was 145.02, 434.75, and 1383.78 s, respectively. The computing time is inherently linked to the available hardware resources. As the proposed algorithm is implemented using MATLAB and the Gurobi solver, the computation time is primarily determined by the solver's efficiency in handling the MILP, which, in turn, depends on both the solver's performance and the hardware resources. As the number of EVs increases, the computation time will naturally extend. However, with more EVs, the available energy for V2G scheduling also increases, potentially leading to higher profits.

V. Hardware Experiment

Experimental validation of the proposed control method is impractical due to the need for a controlled distribution grid and numerous EVs, making it both costly and unfeasible. To address these challenges, a laboratory-level hardware experiment system is employed to evaluate the efficacy of the proposed algorithm, as depicted in Fig. 9. In this system, energy flow is represented by solid orange lines, information flow by dashed blue lines, and the capability for bidirectional communication is depicted with bidirectional arrows. The optimization process starts with the grid sending electricity prices and peak demand charges to the V2G operator, while chargers provide information about current vehicles and their charging targets. After the V2G operator performs the optimization calculations, the results are sent back to the grid for adjustments, and the optimized control signals are transmitted via CAN to the chargers, which manage the EV batteries' charging and discharging. Upon receiving these signals, the chargers adjust the energy flow between the EVs and the grid for effective battery management.

The CMPC-based optimization implemented within the hardware experiment system utilizes a Chroma 17 020 Battery Test System, a ZLG USBCAN-4E-U CAN Bus Analyzer, a control operator/computer, and EV units. The Chroma 17 020 battery test system incorporates bidirectional converters for ac/dc and dc/dc, enabling V2G functionality while facilitating the recording and transmission of charging and discharging data. Thus, within the experiment control system, the Chroma 17 020 battery test system serves as both an actuator and a sensor. The operator, serving as the system's controller, receives data on electricity prices from the grid and battery parameters from EVs. It then executes the proposed CMPC algorithm using MATLAB, with YALMIP interfacing and the Gurobi solver for V2G scheduling optimization, to generate optimized control signals. These signals are relayed to the Chroma 17 020 via the ZLG USBCAN-4E-U CAN Bus Analyzer using the CAN protocol, enabling precise control over battery charging and discharging. Given that communication between chargers and EVs currently relies on CAN signals, leveraging CAN signals for transmitting control signals in the experiment system ensures seamless integration of the proposed CMPC scheduling algorithms into reality. This approach also effectively validates the feasibility of the proposed CMPC algorithm.

The comparison of real and simulated powers for V2G, depicted in Fig. 10, indicates that real power follows simulated

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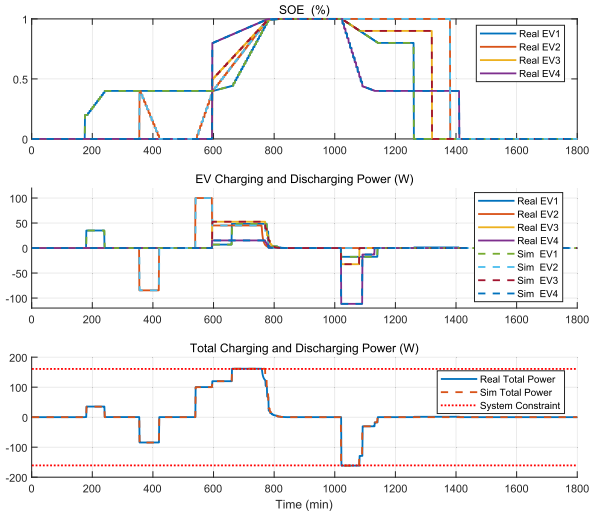

Fig. 10. Experimental result of V2G scheduling optimization.

power with good accuracy. It is worth noting that in this study, four batteries were used to simulate four EVs in the hardware experiment. Nonetheless, the experimental results convincingly demonstrate the effective performance of the proposed algorithm for V2G scheduling optimization. The experimental tests are designed to showcase the real-time performance and effectiveness of the proposed algorithm, where the computer serves as the upper controller, optimizing the charging and discharging process. The algorithm operates in a real-time rolling optimization mode, dynamically incorporating newly arrived EVs into the optimization sequence. However, the current setup is limited to four charging and discharging channels, future research will focus on expanding the number of channels to further validate the algorithm's scalability.

VI. Conclusion

This article proposes a CMPC scheduling method for online V2G optimization to alleviate peak demand charges, range anxiety, and battery degradation, and reduce the computational burden. By constructing a penalty function in the MILP optimization model, EVs can be coordinated to avoid high peak demand charges and reduce frequent power fluctuations. To mitigate battery degradation concerns, a comprehensive aging model that considers both calendar and cycle aging, along with the standard CC–CV charging mode, is used. Range anxiety is addressed by designing constraints of the charging time threshold and SOE threshold. Finally, the control package technology reduces computational and communication burdens, while the rolling optimization method addresses the stochastic arrivals of charging demands. Simulation and hardware experimental results have successfully illustrated the performance of the proposed V2G scheduling scheme. Future research could build upon this foundation by incorporating vehicle communication and routing plan information. By conducting optimal charging route planning and incorporating charging load prediction, a globally optimal solution can potentially be achieved. Furthermore, future research will focus on using charging station data to identify different EVs and enhance the accuracy of their battery-aging models, moving beyond the current method of differentiation based solely on battery capacity and charging/discharging power limits.

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Chuan-Fan Lu (Graduate Student Member, IEEE) received the B.Eng. and M.Eng. degrees from Hainan University, Hainan, China, in 2015 and 2023, respectively. He is currently pursuing the Ph.D. degree with the Center for Control Science and Technology, Southern University of Science and Technology, Shenzhen, China.

His research interests include the intersection of control and optimization with applications in large-scale vehicle-to-grid systems.

Guo-Ping Liu (Fellow, IEEE) received the B.Eng. degree in industrial automation and the M.Eng. degree in control engineering from the Central South University of Technology, Changsha, China, in 1982 and 1985, respectively, and the Ph.D. degree in control systems from the University of Manchester, Manchester, U.K., in 1992.

He is a Professor with the Southern University of Science and Technology, Shenzhen, China. He has authored or co-authored ten books and over 400 journal articles on control systems. His current research interests include the Internet of Things for renewable energy integration and advanced control of industrial systems.

Dr. Liu was a member of the Academy of Europe and a fellow of IET and CAA. He was the General Chair of the 2007 IEEE International Conference on Networking, Sensing and Control, the 2011 International Conference on Intelligent Control and Information Processing, and the 2012 UKACC International Conference on Control. He served as the Editor-in-Chief for *International Journal of Automation and Computing* from 2004 to 2021.

Yi Yu (Member, IEEE) received the M.S. degree in control theory from the University of Electronic Science and Technology of China, Chengdu, China, in 2019, and the Ph.D. degree in control engineering from Wuhan University, Wuhan, China, in 2022.

He is currently a Post-Doctoral Fellow with the Southern University of Science and Technology, Shenzhen, China. His research interests include renewable energy integration, microgrid modeling and control, and networked control systems.

Dr. Yu is a reviewer for many international journals.

Jinqiang Cui received the B.E. and M.Sc. degrees in mechatronic engineering from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2008, respectively, and the Ph.D. degree in electrical and computer engineering from the National University of Singapore, Singapore, in 2015.

He was a Sensor Design Engineer with HSG-IMIT, Villingen-Schwenningen, Germany, from 2008 to 2010, and served as the Chief Technology Officer for a UAV Start-Up company, from 2016 to 2019, before joining the Pengcheng Laboratory, Shenzhen, China. His current research interests include embodied artificial intelligence, simultaneous localization and mapping, and intelligent multirobot systems.

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